

Assignment 3 Instructions

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ASSIGNMENT: DYNAMIC INPUT FOR CONTROL FLOWS AND LOOPS

Objective

Modify the starter.html and JavaScript to accept **dynamic user input** instead of using hardcoded values. The assignment covers two key concepts:

1. **If-Else Statements** - Accept a number input and determine if it is positive, negative, or zero.
2. **Switch Statements** - Accept a day number (0-6) and output the corresponding day name.

Expected Functionality

1. Positive/Negative Number (If-Else)

- An input field should accept a numeric value.
- When the “Run” button is clicked, the JavaScript should:
- If the number > 0 : display “The number X is positive.”
- If the number < 0 : display “The number X is negative.”
- If the number $== 0$: display “The number is zero.”

2. Day of the Week (Switch Statement)

- An input field should accept a number between 0 and 6.
- When the “Run” button is clicked, the JavaScript should:
- Use a switch statement to map the number to a day name:
- 0 $\rightarrow$ Sunday, 1 $\rightarrow$ Monday, 2 $\rightarrow$ Tuesday, 3 $\rightarrow$ Wednesday, 4 $\rightarrow$ Thursday, 5 $\rightarrow$ Friday, 6 $\rightarrow$ Saturday
- If the input is outside 0-6, display “Invalid day.”

Hints & Tips

Selecting Input Values

Use `document.getElementById('input-id').value` to get the value from an `<input>` element. The value will be a string, so you'll need to convert it to a number:

```
const num = Number(document.getElementById('num-input').value);
```

If-Else Logic

Structure your conditionals similar to the existing runIfElse() function, but use the dynamic number value instead of the hardcoded num = 7.

Switch Statement

Structure your switch similar to the existing runSwitch() function, but use the dynamic day value. Remember to handle invalid inputs (numbers outside 0-6) with a default case.

Converting String to Number

JavaScript's Number() function converts a string to a numeric value. If the input is not a valid number, the result will be NaN. You can check for this:

```
if (isNaN(num)) {  
    output.innerText = 'Please enter a valid number.';  
} else {  
    // proceed with your logic  
}
```

Submission

Create a new file called dynamic_input.html with the input boxes and update the JavaScript functions to use dynamic input. Also create an Assignment Steps.md file describing the task and providing hints as shown above.